

B.TECH DATA ANALYTICS — FINAL PROJECT REPORT

Electricity Consumption & Demand Forecasting: EDA, Visualization and Executive Story

EDA, Visualization and Executive Story

Domain: Energy / Electricity Analytics

Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Technology in Data Analytics

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Table of Contents

Abstract / Executive Summary

This report presents the final documentation of a B.Tech Data Analytics project titled "Electricity Consumption & Demand Forecasting: EDA, Visualization and Executive Story." The project analyzes a synthetic electricity consumption dataset to explore consumption, peak demand, generation, pricing, and renewable-energy patterns, and applies a complete analytics workflow spanning data cleaning, exploratory data analysis, statistical hypothesis testing, machine learning, time-series forecasting, SQL-based querying, and Power BI dashboard development.

A Welch's two-sample t-test confirmed a statistically significant difference in electricity consumption between peak and non-peak periods ($p < 0.001$). Correlation analysis showed that Consumption_MWh and Generation_MWh have a moderate positive relationship with Peak_Demand_MW, while Electricity Price shows a weaker relationship, and Weather Index and Renewable Share show negligible linear association. A Random Forest regression model and an ARIMA(1,1,1) forecasting model were developed following chronological validation practices to prevent data leakage; their exact performance metrics are pending final notebook execution and are marked accordingly in this report.

Note: The dataset used in this project is synthetic and was generated for academic and practice purposes. It is not official U.S. Energy Information Administration (EIA) or utility data, and no findings in this report should be interpreted as representing real-world electricity markets.

Introduction

Electricity systems generate large volumes of operational data — consumption, generation, demand, pricing, and renewable-mix information — that can be analyzed to improve grid reliability, cost efficiency, and sustainability planning. This project was undertaken to apply modern data analytics techniques to such data using a structured, end-to-end pipeline.

The report is organized to follow the natural progression of an analytics project: starting with the business context and dataset, moving through data quality assessment and cleaning, exploratory and statistical analysis, predictive modelling (machine learning and time-series forecasting), and concluding with SQL-based reporting, dashboard design, and business recommendations. Each section presents only the information and results supplied for this project; sections requiring metrics not yet computed are clearly marked [TO BE ADDED].

Business Problem

Electric utilities and grid operators must continuously balance electricity supply (generation) against demand (consumption), while managing peak-load events, integrating renewable energy sources, and responding to price volatility. Without clear visibility into consumption drivers and peak-demand behaviour, utilities risk inefficient capacity planning, higher operating costs, and reduced grid reliability.

This project addresses the question: how can historical electricity consumption, generation, demand, price, and renewable-share data be analyzed to understand peak-demand behaviour and support capacity planning and short-term forecasting?

Project Objectives

- Perform a structured data quality audit and cleaning process on the raw electricity dataset.
- Conduct exploratory data analysis of consumption, demand, generation, price, and renewable-share patterns across states, sectors, and time.
- Statistically test whether peak-flagged periods differ significantly in consumption from non-peak periods.
- Build a machine learning model to estimate Peak Demand from operational and environmental features.
- Develop a time-series forecasting model for daily average peak demand.
- Perform SQL-based analytical querying for KPI reporting.
- Design a two-page Power BI dashboard for executive monitoring and analysis.
- Document all findings, assumptions, and limitations transparently and reproducibly.

Dataset Description

The dataset used in this project is a synthetic electricity consumption dataset, generated specifically for academic and practice purposes. It is explicitly not sourced from the U.S. Energy Information Administration (EIA) or any official government or utility body.

Table 1: Dataset summary

Attribute	Value
Dataset type	Synthetic (academic/practice use only)
Original size	5,045 rows × 11 columns
Final cleaned size	4,991 rows × 11 columns
Domain	Energy / Electricity Analytics

Data Dictionary

Table 2: Column definitions

Column	Description
Record_ID	Unique identifier for each record
Date_Time	Timestamp of the recorded observation
State	U.S. state associated with the record
Sector	Consumption sector (e.g., residential, commercial, industrial)
Consumption_MWh	Electricity consumption in megawatt-hours
Peak_Demand_MW	Peak electricity demand in megawatts
Generation_MWh	Electricity generation in megawatt-hours
Electricity_Price_USD_per_MWh	Electricity price in USD per megawatt-hour
Renewable_Share_Pct	Percentage share of renewable energy in generation
Weather_Index	Index representing weather-related conditions
Peak_Flag	Categorical flag (Yes/No) indicating a peak demand period

Data Acquisition & Provenance

The dataset was obtained as a pre-generated synthetic dataset designed to simulate realistic electricity consumption and demand patterns for academic learning purposes.

- Source type: Synthetic / generated dataset — not official EIA or utility data.
- Format: Structured tabular data (CSV) with 11 columns.
- Coverage: Multiple U.S. states and consumption sectors across a defined date range.

Note: Because the dataset is synthetic, all findings and model results in this report are illustrative of methodology rather than representative of real-world U.S. electricity markets.

Data Audit & Quality Assessment

A structured data quality audit was performed prior to cleaning to establish a baseline understanding of missingness, duplication, and validity issues in the raw dataset (5,045 rows).

Missing Value Audit

Table 3: Non-null counts per column (out of 5,045 original rows)

Column	Non-Null Count	Missing Count
State	4,919	126
Sector	4,943	102
Consumption_MWh	4,955	90
Peak_Demand_MW	4,970	75
Generation_MWh	4,943	102
Electricity_Price_USD_per_MWh	4,935	110
Renewable_Share_Pct	4,954	91
Weather_Index	4,919	126

Duplicate and Validity Checks

Table 4: Data quality issues identified

Check	Count Found
Exact duplicate rows	25
Invalid dates	8
Negative Consumption_MWh	8
Negative Generation_MWh	8
Invalid Renewable Share values	5
Negative Peak Demand	0
Negative Electricity Price	8

Note: Negative electricity prices were retained rather than removed, since negative prices are a documented, real phenomenon in electricity markets (e.g., during periods of oversupply).

Statistical Outlier Analysis (IQR Method)

Table 5: Outliers detected using the interquartile range (IQR) method

Variable	Outliers Detected
Consumption_MWh	21
Peak_Demand_MW	47
Generation_MWh	34

Variable	Outliers Detected
Electricity_Price_USD_per_MWh	51
Renewable_Share_Pct	19
Weather_Index	27

Note: Statistical outliers were not automatically removed, because an outlier is not necessarily an invalid observation. Values were retained pending further contextual review.

Data Cleaning & Preprocessing

The following cleaning and preprocessing operations were applied to produce the final, analysis-ready dataset:

1. Missing-value audit — quantified missingness across all columns.
2. Duplicate removal — 25 exact duplicate rows identified and removed.
3. State standardization — normalized inconsistent state naming/formatting.
4. Sector standardization — normalized inconsistent sector labels.
5. Date conversion — converted Date_Time to a proper datetime format.
6. Invalid date removal — 8 invalid date entries handled.
7. Missing categorical-value handling — addressed missing State/Sector values.
8. Missing numerical-value handling — addressed missing values in numeric columns.
9. Invalid numeric-value handling — reviewed negative consumption, generation, and price entries, and invalid renewable-share values.
10. Final validation — confirmed the cleaned dataset was consistent and ready for analysis.

Table 6: Result of the cleaning process

Stage	Rows	Columns
Original dataset	5,045	11
Final cleaned dataset	4,991	11

Exploratory Data Analysis

Exploratory data analysis was conducted on the cleaned dataset to understand distributions, relationships, and temporal patterns across electricity consumption, generation, demand, price, and renewable share.

Analyses Performed

- Descriptive statistics
- State, Sector, and Peak Flag distribution
- Date range, monthly, daily, hourly, and day-of-week analysis
- Correlation analysis
- Histograms and scatter plots
- State-wise and sector-wise comparisons
- Peak vs non-peak analysis
- Renewable-share and electricity-price analysis
- Generation vs consumption analysis
- Top consumption, peak-demand, and generation records
- Generation/Consumption ratio
- 7-day moving average
- Seasonality analysis and seasonal decomposition

Peak Flag Distribution

Table 7: Distribution of Peak_Flag in the cleaned dataset

Peak Flag	Record Count
No	4,284
Yes	753

Peak-flagged records represent a minority class (approximately 15% of the cleaned dataset), consistent with the expectation that peak-demand periods occur less frequently than normal operating periods.

Correlation with Peak Demand

Table 8: Pearson correlation of numeric variables with Peak_Demand_MW

Variable	Correlation with Peak_Demand_MW
Consumption_MWh	0.650441
Generation_MWh	0.644786
Electricity_Price_USD_per_MWh	0.188429
Weather_Index	0.005092
Record_ID	-0.005924
Renewable_Share_Pct	-0.011491

Consumption and Generation show a moderate positive correlation with Peak Demand, while Electricity Price shows a weaker positive relationship. Weather Index and Renewable Share show negligible linear correlation with Peak Demand in this dataset.

EDA Visuals

[INSERT EDA VISUAL HERE]

Figure: Distribution and correlation visuals from the EDA notebook (histograms, scatter plots, moving average, seasonal decomposition).

Statistical Analysis

Research Question

"Do Peak Flag = Yes records have significantly higher electricity consumption than Peak Flag = No records?"

Hypotheses

- H0 (Null Hypothesis): There is no significant difference in mean electricity consumption between peak and non-peak periods.
- H1 (Alternative Hypothesis): There is a significant difference in mean electricity consumption between peak and non-peak periods.

Method

Welch's independent two-sample t-test was used, as it does not assume equal variances between groups, making it appropriate for comparing peak and non-peak consumption groups of unequal size.

Results

Table 9: Welch's t-test results for peak vs non-peak consumption

Metric	Value
Non-Peak mean consumption	907.413805 MWh
Peak mean consumption	1,392.147797 MWh
Mean difference	484.7339919 MWh
t-statistic	63.67779008
p-value	$p < 0.001$
95% Confidence Interval	469.8032918 to 499.6646919 MWh

Decision & Interpretation

Since the p-value is far below the conventional significance threshold of 0.05, the null hypothesis (H0) is rejected. Peak periods are associated with substantially higher electricity consumption than non-peak periods.

Note: This is a statistically significant association, not a causal relationship — peak periods are not stated to cause higher consumption. Additionally, Peak_Flag may itself be derived from Peak_Demand_MW (which is correlated with Consumption_MWh), so this relationship should be interpreted carefully.

Statistical Test Visual

[INSERT STATISTICAL TEST RESULT VISUAL HERE]

Figure: Distribution comparison of consumption between peak and non-peak groups.

Machine Learning

Objective

A supervised regression model was built to estimate Peak_Demand_MW from operational and environmental features.

Target and Features

Table 10: Machine learning setup

Role	Variable(s)
Target	Peak_Demand_MW
Features	Consumption_MWh, Generation_MWh, Electricity_Price_USD_per_MWh, Renewable_Share_Pct, Weather_Index
Excluded	Peak_Flag (excluded — potential leakage/derivation from Peak_Demand_MW)

Train/Test Strategy

- Chronological 80/20 split (shuffle=False) to preserve temporal order and avoid look-ahead bias.

Baseline and Model

Table 11: Model configuration

Component	Detail
Baseline	DummyRegressor (mean strategy)
Model	RandomForestRegressor
n_estimators	100
random_state	42

Evaluation Metrics

Table 12: Baseline vs. model performance

Metric	Baseline (DummyRegressor)	Random Forest Model
MAE	[TO BE ADDED]	[TO BE ADDED]
RMSE	[TO BE ADDED]	[TO BE ADDED]
R ²	[TO BE ADDED]	[TO BE ADDED]

Feature importance and actual-vs-predicted analysis were performed to interpret model behaviour and validate prediction quality.

Model Diagnostics Visual

[INSERT ML ACTUAL VS PREDICTED PLOT HERE]

Figure: Actual vs. predicted Peak_Demand_MW and feature importance ranking from the Random Forest model.

Note: Actual MAE, RMSE, and R^2 values are not fabricated in this report. They must be filled in from the executed notebook output before final submission.

Time-Series Forecasting

Objective

Daily average Peak Demand was forecasted using a classical time-series model to support short-term operational planning.

Workflow

- 11. Aggregate Peak_Demand_MW by day to produce a daily average series.
- 12. Apply a chronological train/test split to preserve temporal ordering.
- 13. Establish a baseline forecast for comparison.
- 14. Fit an ARIMA model on the training series.
- 15. Evaluate forecast accuracy on the held-out test period.
- 16. Compare ARIMA performance against the baseline.

Model Specification

Table 13: Forecasting configuration

Parameter	Value
Model	ARIMA(1,1,1)
Validation approach	Chronological (no shuffling)

Evaluation

Table 14: Baseline vs. ARIMA forecast performance

Metric	Baseline	ARIMA(1,1,1)
MAE	[TO BE ADDED]	[TO BE ADDED]
RMSE	[TO BE ADDED]	[TO BE ADDED]

Forecast Visual



Figure: ARIMA forecast vs. actual daily average Peak Demand, compared against the baseline.

Note: Actual ARIMA evaluation metrics are not fabricated in this report. They must be filled in from executed notebook output before final submission.

SQL Analysis

Database Setup

Table 15: SQL database configuration

Item	Value
Database	electricity_project
Table	energy_consumption
Imported records	4,991

Queries Completed

- Overall KPIs (total consumption, generation, average peak demand, etc.)
- State-wise average consumption
- State-wise average peak demand
- Sector-wise average consumption
- Peak vs non-peak comparison
- Monthly consumption
- Monthly peak demand
- Top 10 peak demand records
- State average electricity price
- State average renewable share
- Top 5 states by total consumption

Sample Query Structure

```
SELECT State,
       AVG(Consumption_MWh) AS Avg_Consumption
FROM energy_consumption
GROUP BY State
ORDER BY Avg_Consumption DESC;
```

Note: The complete SQL scripts are maintained in the project repository (sql/electricity_analysis.sql). Query result values are not reproduced here; refer to the executed SQL output for specific figures.

Power BI Dashboard

The Power BI report consists of two pages designed to serve both executive monitoring and detailed analytical needs.

Page 1 — Energy Analytics Dashboard

KPI Cards

- Total Consumption
- Total Generation
- Average Peak Demand
- Average Renewable Share

Slicers

- Date
- State
- Sector
- Peak_Flag

Visuals

- Average Electricity Consumption by State
- Monthly Consumption vs Peak Demand
- Peak vs Non-Peak Consumption
- Average Renewable Energy Share by State
- Generation vs Consumption
- Average Electricity Price by State
- Consumption Share by Sector

[INSERT POWER BI PAGE 1 SCREENSHOT HERE]

Figure: Energy Analytics Dashboard (Page 1).

Page 2 — Energy Analysis Report

- Daily Electricity Consumption Trend
- Daily Peak Demand Trend
- Average Peak Demand by State
- Monthly Generation Trend
- Consumption Share by Sector
- Average Renewable Share Gauge

[INSERT POWER BI PAGE 2 SCREENSHOT HERE]

Figure: Energy Analysis Report (Page 2).

Key Business Insights

The table below distinguishes between the observed (measured) result, the business interpretation of that result, and any related recommendation. Interpretations and recommendations are clearly separated from the underlying observational findings.

Table 16: Observed results, interpretations, and recommendations

Observed Result	Business Interpretation	Recommendation
Peak-flagged periods show significantly higher mean consumption (1,392.15 MWh) than non-peak periods (907.41 MWh), $p < 0.001$.	Peak periods are associated with, but not proven to cause, substantially higher consumption.	Continue monitoring peak-flagged periods as an indicator of elevated consumption.
Consumption_MWh correlates moderately with Peak_Demand_MW ($r = 0.65$).	Higher consumption tends to coincide with higher peak demand.	Track consumption trends as an early input for peak-demand monitoring.
Generation_MWh correlates moderately with Peak_Demand_MW ($r = 0.64$).	Higher generation activity tends to coincide with higher peak demand.	Align generation planning with observed demand trends.
Electricity_Price_USD_per_MWh shows a weaker positive correlation with Peak_Demand_MW ($r = 0.19$).	Price movements are less strongly associated with peak demand than consumption or generation.	Treat price as a secondary, not primary, peak-demand indicator.
Weather_Index ($r = 0.005$) and Renewable_Share_Pct ($r = -0.011$) show negligible linear correlation with Peak_Demand_MW.	No strong linear relationship was found between these variables and peak demand in this dataset.	Investigate potential non-linear effects before drawing further conclusions.
The dataset contains state, sector, and temporal variation.	This structure supports segmentation for monitoring purposes.	Use state/sector/time segmentation in dashboards for operational monitoring.

Note: All interpretations above describe observed associations only. None of the findings in this report should be read as establishing a causal relationship.

Business Recommendations

The following recommendations are proposed based on the measured findings in this report. They are suggested directions for further evaluation, not proven outcomes, and should be validated with domain expertise and real operational data before implementation.

Peak-Demand Monitoring

- Continue tracking Peak_Flag periods against Consumption_MWh and Generation_MWh, given their observed positive association with Peak_Demand_MW.

Capacity Planning

- Use the historical peak vs non-peak consumption gap as one input (not the sole determinant) for capacity buffer planning.

Demand-Side Management

- Explore demand-side management programs targeted at periods historically associated with peak flags, subject to further investigation.

Energy Generation Planning

- Align generation planning with observed consumption trends, given the moderate correlation between Generation_MWh and Peak_Demand_MW.

Renewable Energy Monitoring

- Continue monitoring Renewable_Share_Pct across states and sectors, even though it shows weak linear correlation with peak demand in this dataset.

Forecasting Support

- Use the ARIMA-based daily peak-demand forecast as a supplementary planning input, recognizing its dependence on historical patterns.

Power BI-Based Operational Monitoring

- Adopt the two-page Power BI dashboard for ongoing operational and executive-level monitoring of KPIs, state/sector trends, and peak behaviour.

Assumptions

- The dataset is synthetic and was generated for academic/practice purposes.
- Negative electricity prices were retained, since negative prices can occur in real electricity markets.
- Weather_Index has no documented valid range.
- Statistical outliers were not automatically removed.
- Peak_Flag may be derived from Peak_Demand_MW.
- Historical patterns may not always continue in future periods.
- Model performance depends on the available dataset and selected features.

Limitations

- The synthetic dataset limits generalization to real-world electricity markets.
- No official operational data was used in this project.
- Some data-quality issues required cleaning and imputation, which may introduce minor distortions.
- Correlation does not imply causation; no causal relationships are established in this report.
- Forecasting uncertainty exists, and ARIMA(1,1,1) is a relatively simple model.
- The derivation of Peak_Flag from Peak_Demand_MW is not fully documented.
- Weather_Index interpretation is limited by the absence of a documented scale.
- Machine learning and forecasting performance should be evaluated strictly within the context of this dataset.

Privacy & Responsible Analytics

- The dataset does not contain personal or sensitive information; all records pertain to aggregated state/sector-level energy metrics.
- Data leakage was considered during model design.
- Peak_Flag was excluded from the machine learning feature set because of potential leakage/derivation concerns.
- Chronological validation was used for time-series modelling to avoid look-ahead bias.
- Statistical uncertainty (confidence intervals, p-values) is communicated rather than presented as absolute certainty.
- No causal claims are made from observational or correlational relationships.
- Results are intended to be reproducible through documented code and a transparent workflow.

Validation & Reproducibility

Validation Approach

- Chronological (shuffle=False) 80/20 train/test split used for the machine learning model.
- Chronological train/test split used for ARIMA time-series forecasting.
- Baseline models (DummyRegressor for ML; a naive baseline for forecasting) used to contextualize model performance.

Reproducibility Measures

- Fixed random_state = 42 used for the RandomForestRegressor.
- Documented, step-by-step data-cleaning and preprocessing pipeline.
- Modular notebook structure separating cleaning, EDA, statistical analysis, ML, and forecasting.
- SQL scripts version-controlled in the project repository for repeatable KPI querying.

Conclusion

This project demonstrates a complete, end-to-end data analytics workflow applied to a synthetic electricity consumption dataset — spanning data auditing, cleaning, exploratory analysis, statistical hypothesis testing, machine learning, time-series forecasting, SQL analysis, and Power BI dashboard development.

The statistical analysis confirmed a significant difference in consumption between peak and non-peak periods, while exploratory analysis revealed moderate positive correlations between consumption, generation, and peak demand. Machine learning and time-series forecasting components were implemented using sound methodological practices (chronological validation, baseline comparison, leakage avoidance); exact performance metrics are to be finalized from notebook execution outputs.

Overall, the project fulfils its academic objective of demonstrating a rigorous, transparent, and responsible analytics methodology suitable for B.Tech Data Analytics evaluation, while explicitly avoiding unsupported claims, invented metrics, or causal overreach.

References / Tools Used

This project did not draw on external literature or third-party data sources beyond the synthetic dataset described in Section 6. The following tools and libraries were used to complete the analysis:

Table 17: Tools and technologies used

Category	Tool(s)
Programming language	Python
Data handling	pandas, numpy
Visualization	matplotlib, seaborn
Statistical analysis	scipy / statsmodels (Welch's t-test)
Machine learning	scikit-learn (RandomForestRegressor, DummyRegressor)
Time-series forecasting	statsmodels (ARIMA)
Database	MySQL
Dashboarding	Power BI
Development environment	Jupyter Notebook